

Work and Entanglement in Coupled Quantum Oscillators

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Abstract

We extend a classical question—how much work accompanies a reduction in dependence?—to quantum systems. Quantum mutual information is the direct analogue of classical mutual information, but it includes classical as well as quantum correlations. Entanglement therefore requires its own measure. For two-mode Gaussian states, both quantities can be calculated from the covariance matrix, while work is determined by the controlled Hamiltonian and protocol. A two-mode squeezed vacuum gives a compact example in which removing entanglement can release energy. We outline how to measure a work-versus-entanglement curve in a quantum optomechanical oscillator experiment.

1 From a dependence ratio to a quantum state

For classical variables, the dependence ratio is $r(x, y) = p(x, y)/(p_X(x)p_Y(y))$, and its average log is the mutual information. The quantum analogue starts with a joint density operator ρ_{AB} and reduced states $\rho_A = \text{Tr}_B \rho_{AB}$, $\rho_B = \text{Tr}_A \rho_{AB}$. Define

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) = D(\rho_{AB} \| \rho_A \otimes \rho_B), \quad (1)$$

where $S(\rho) = -\text{Tr}(\rho \log \rho)$ and $D(\rho \| \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]$. For states diagonal in a common product basis, this reduces to classical mutual information. In general, there is no pointwise quantum ratio $r(x, y)$: noncommuting operators replace probabilities.

Mutual information is still broader than entanglement. For example, $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ has nonzero mutual information but is separable. To target entanglement, use a measure such as logarithmic negativity

$$E_N(\rho_{AB}) = \log \|\rho_{AB}^{T_B}\|_1, \quad (2)$$

where T_B is partial transpose and $\|\cdot\|_1$ is trace norm. For two-mode Gaussian states, covariance-matrix criteria give an efficient entanglement test and quantification.[1, 2, 3]

2 A solvable oscillator example

Take two equal-frequency bosonic modes with bare Hamiltonian

$$H_0 = \hbar\omega(a^\dagger a + b^\dagger b), \quad (3)$$

and prepare a two-mode squeezed vacuum

$$|\psi_r\rangle = \frac{1}{\cosh r} \sum_{n=0}^{\infty} (\tanh r)^n |n, n\rangle. \quad (4)$$

The squeezing parameter r controls both correlations and energy. Each mode has mean occupation $\bar{n} = \sinh^2 r$. The state is pure, so its quantum mutual information is twice the entropy of either mode:

$$I(A:B) = 2g(\bar{n}), \quad g(n) = (n+1) \log(n+1) - n \log n. \quad (5)$$

Its logarithmic negativity is $E_N = 2r$ (natural-log convention), and its energy above the two-mode vacuum is

$$\Delta E = \langle H_0 \rangle = 2\hbar\omega \sinh^2 r. \quad (6)$$

Thus I and E_N are related along this particular state family, but they are different measures and need not track one another for mixed states.

For a closed system driven by a time-dependent control Hamiltonian $H(t)$, the work done on the system is

$$W_{\text{on}} = \int_{t_0}^{t_1} \text{Tr}[\rho(t) \partial_t H(t)] dt. \quad (7)$$

If the control interaction is off at both endpoints, $W_{\text{on}} = \Delta \langle H_0 \rangle$. An ideal global unsqueezing operation taking r_0 to r_1 therefore gives

$$W_{\text{on}} = 2\hbar\omega (\sinh^2 r_1 - \sinh^2 r_0). \quad (8)$$

For $r_0 = 0.5 \rightarrow r_1 = 0$, E_N falls from 1 nat to zero, while $W_{\text{on}} = -0.543 \hbar\omega$: the ideal process extracts that energy. At $\omega/2\pi = 400$ kHz, this is 1.44×10^{-28} J extracted. This is not a universal cost of disentanglement. It is the energy released from this specified initial state under a global, reversible control; other states, Hamiltonians, baths, and allowed operations give different work-versus-entanglement curves. Thermodynamic work and quantum correlations have been connected in several operational settings.[4]

3 A route to an experiment

A suitable platform is a pair of quantum-regime mechanical modes coupled through a microwave or optical cavity. Experiments have demonstrated stabilized entanglement between massive mechanical oscillators, inferred from correlated mechanical fluctuations and cavity readout.[5] For a two-mode Gaussian state, reconstruct the quadrature covariance matrix V from calibrated measurements. Its symplectic eigenvalues give $S(\rho_{AB})$ and those of the reduced modes give $S(\rho_A), S(\rho_B)$, hence $I(A:B)$. Partial transpose flips the sign of one momentum quadrature; if its smallest symplectic eigenvalue is $\tilde{\nu}_- < \hbar/2$, the state is entangled, with

$$E_N = \max \left\{ 0, -\log \left(\frac{2\tilde{\nu}_-}{\hbar} \right) \right\}. \quad (9)$$

Next, vary a calibrated control parameter (for example, a parametric drive or cavity-mediated coupling) and record the state covariance, pump work, and heat flow at each setting. The resulting empirical curve W_{on} versus ΔE_N is the quantum counterpart of our original work-versus-dependence question. A strong test would compare measured work with a model fitted to the same Hamiltonian and damping, and repeat the protocol at several temperatures and rates. The point is not to assume a universal $k_B T \Delta E_N$ law, but to find which parts of the curve follow from the oscillator model and which arise from the chosen control and environment.

4 Preliminary match to public data

A useful first check is the public dataset for entanglement of propagating optical modes mediated by a mechanical resonator. The experiment reports the Duan-Giedke-Cirac-Zoller inseparability witness $\mathcal{I} = 0.83$ (with 2% statistical and 0.3% systematic uncertainty) and the smallest partially transposed symplectic eigenvalue $2\tilde{\nu}_- = 0.79$. Its reported logarithmic negativity is $E_N = 0.35$ bits.[6]

For this reported normalization, the Gaussian-state formula in Section 3 becomes

$$E_N = \max \left\{ 0, -\log \left(\frac{2\tilde{\nu}_-}{\hbar} \right) \right\}, \quad \frac{2\tilde{\nu}_-}{\hbar} = 0.79. \quad (10)$$

Substitution gives the calculation explicitly:

$$E_N^{(\text{nats})} = \max\{0, -\ln(0.79)\} = 0.2357, \quad E_N^{(\text{bits})} = \frac{E_N^{(\text{nats})}}{\ln 2} = \frac{0.2357}{0.6931} = 0.340 \text{ bits}. \quad (11)$$

The reported value is 0.35 bits, a difference of 0.010 bits (about 3%) from this estimate. Since 0.79 is rounded to two decimals, its underlying value lies approximately in $[0.785, 0.795]$, which maps to $E_N \in (0.330, 0.349]$ bits; the reported value is consistent at the precision of the published rounded input. Independently, the witness gives $\mathcal{I} = 0.83 < 1$, on the entangled side of its separability threshold. This is a direct preliminary check of the state-analysis formalism.

Two further arithmetic checks quantify the operating point. The paper’s reduced efficiency model gives

$$\mathcal{I}_{\min} = 1 - \frac{\eta_{\text{meas}}}{2} = 1 - \frac{0.58}{2} = 0.71; \quad \mathcal{I}_{\text{obs}} - \mathcal{I}_{\min} = 0.83 - 0.71 = 0.12. \quad (12)$$

The fuller optomechanical model includes device details and agrees with the phase-dependent data. For the reported drive-induced quantum-backaction rates and thermal decoherence rate, the common 2π factor cancels:

$$\frac{\Gamma_{\text{qba}}}{\Gamma_{\text{th}}} = \frac{2\pi(1.35 + 0.89) \text{ kHz}}{2\pi(0.20) \text{ kHz}} = \frac{2.24}{0.20} = 11.2. \quad (13)$$

This ratio is a control-strength proxy, not an energy or a work estimate. In the paper’s work formalism, $W_{\text{on}} = \int \text{Tr}[\rho(t) \partial_t H(t)] dt$; the published rates and single operating-point covariance do not supply the calibrated $H(t)$ and state trajectory needed to evaluate that integral. Thus this dataset checks the entanglement calculation and supplies a quantitative operating-point benchmark, while a work-versus-entanglement curve requires calibrated drive power and losses across multiple settings. The public archive contains the raw and source data for the figures, enabling frequency-resolved covariance and entanglement estimates as a next calculation.[6]

References

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